Directional and Void-Conditioned Variations in the Local Hubble Relation

Evidence for Environment-Structured Low-Redshift Hubble Anisotropy

James A. Darke

March 8, 2026

Abstract

We test whether the low-redshift Hubble relation is direction-dependent and whether any such dependence is conditioned by line-of-sight environment. Using the Pantheon+SH0ES distance table as the supernova backbone, we construct a sequence of increasingly strict analyses. We first identify a large-angle anisotropy candidate in distance-modulus residuals, then fit a dipolar residual field directly, and then test whether the inferred dipole is structured by independent measures of local environment. The analysis proceeds through hemisphere scans, dipole fitting, shell and cumulative stability checks, custom environment and stratification proxies, matched and rotated null tests, control-axis comparisons, a failed CF4/EDD replication, and a successful published-void replication.

For the reference sample over $0.01 \leq z_{\mathrm{CMB}} \leq 0.15$, the fitted dipole amplitude is 0.0808 mag, with axis $(\mathrm{RA}, \mathrm{Dec}) = (23.3^\circ, 37.6^\circ)$. The dipole survives shuffle-based null tests and remains stable under several redshift truncations. A custom galaxy-based stratification field produces strong projection-side and sector-side correlations. Redshift-matched and radial-scramble nulls do not erase the signal, although a crude sky-rotation test shows that the effect is not uniquely distinguishable from general large-angle sky structure if one uses only simple global correlation metrics. This ambiguity is partly resolved by axis-specific tests, which show that the fitted dipole is unusually effective at splitting the supernova sample into opposite sectors with different redshift-matched stratification and residual behaviour. A CF4/EDD-based proxy does not independently replicate the result. However, replication against a published local-void catalogue succeeds: dipole sectors exhibit significantly different overlap with catalogued voids, including in redshift-matched sector splits.

Taken together, the results support a low-redshift picture in which Hubble residual anisotropy is not merely directional, but environment-structured. The strongest evidence is sector-based and redshift-matched, and the most convincing independent replication comes from a published local-void field rather than from the CF4 proxy.

1 Introduction

The local Hubble relation is often compressed into a single scalar parameter, usually denoted H_0. That compression is operationally useful, but it can also obscure directional and environment-dependent structure if late-time expansion inference is sensitive to how different lines of sight sample the nearby cosmic web. The central empirical question of this paper is therefore not whether one can infer a best-fit global value of H_0, but whether low-redshift distance-modulus residuals exhibit structured directional variation, and whether that variation is conditioned by nearby environment.

The present work was motivated by a broader closure-stratification framework in which late-time cosmological inference need not reduce cleanly to a single homogeneous parameter. The aim of the paper, however, is strictly observational. We do not assume that broader framework is correct. Instead, we ask a narrower and more defensible question: does the local Hubble residual field show evidence of large-angle anisotropy that is systematically related to nearby void and wall structure?

To answer this, we built a staged evidence chain. Each stage was designed either to detect signal, to harden it, or to falsify weaker interpretations. A visually suggestive directional pattern is not enough. Any candidate signal must survive survey jackknifes, redshift-floor changes, balance constraints, matched nulls, and independent environment products before it can be treated seriously.

Throughout the paper we distinguish between the conventional late-time Hubble-constant language and the effective quantity actually fit in the present work. The parameter extracted from the intercept-only low-redshift baseline is denoted H_{eff}, since the entire point of the analysis is that the local low-redshift relation may not be fully described by one isotropic scalar.

2 Data and Baseline Construction

2.1 Supernova sample

The analysis uses the Pantheon+SH0ES distance table `Pantheon+SH0ES.dat` as the supernova backbone. We work directly at the distance-product level rather than refitting light curves. The main columns used are:

- `CID` for supernova identity,
- `zCMB` for the CMB-frame redshift,
- `MU_SH0ES` for the distance modulus observable,
- `MU_SH0ES_ERR_DIAG` for first-pass diagonal uncertainty,
- `RA`, `DEC` for sky position,
- `IS_CALIBRATOR` to exclude calibrator objects from the main Hubble fit.

We apply the following selection:

1. exclude calibrators,
2. require valid sky coordinates, redshift, modulus, and diagonal uncertainty,
3. restrict to $0.01 \leq z_{\mathrm{CMB}} \leq 0.15$ for the reference sample,
4. deduplicate repeated supernova identifiers by retaining the row with the smallest diagonal modulus uncertainty.

The lower redshift cut is deliberately conservative so that peculiar velocities do not fully dominate the first-pass directional tests.

2.2 Baseline Hubble model

For the initial exploratory analysis we use the intercept-only low-redshift baseline

$$\mu_{\text{model}} = 5\log_{10}(c\, z_{\text{CMB}}) + A, \tag{1}$$

where A is fit by weighted least squares and is related to an effective Hubble parameter by

$$H_{\text{eff}} = 10^{(25-A)/5}. \tag{2}$$

This is not presented as the final cosmographic model. It is a deliberately simple baseline used to identify directional structure in residual space:

$$\Delta\mu = \mu_{\text{obs}} - \mu_{\text{model}}. \tag{3}$$

The reference fit gives

$$A = 15.785349 \pm 0.008423, \qquad H_{\text{eff}} \approx 69.651\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}.$$

2.3 Dipole model

After the initial hemisphere scan, the residual field is fit directly with a dipolar model

$$\Delta\mu(\hat{\mathbf{n}}) = a + \mathbf{b}\cdot\hat{\mathbf{n}}, \tag{4}$$

where $\hat{\mathbf{n}}$ is the unit vector to the supernova. The dipole amplitude is

$$|\mathbf{b}| = \sqrt{b_x^2 + b_y^2 + b_z^2}. \tag{5}$$

For the reference sample the fitted dipole amplitude is

$$|\mathbf{b}| = 0.080799\ \text{mag},$$

with axis

$$(\text{RA}, \text{Dec}) = (23.331^\circ,\ 37.617^\circ).$$

3 Methodological Roadmap

The full analysis was built in nine stages. This section gives a concise roadmap so that the later results can be read as one evidentiary sequence rather than as disconnected tests.

- **Stage 1:** hemisphere scans to identify large-angle anisotropy in the residual field.
- **Stage 2:** survey jackknifes, redshift-floor sweeps, shell splits, and balance-aware scans.
- **Stage 3:** direct dipole fitting and radial stability checks.
- **Stages 4–5:** custom galaxy-based environment and stratification fields.

- **Stage 6:** redshift-matched, sky-rotation, and radial-scramble falsification tests.
- **Stage 7:** control-axis specificity tests against random and mask-matched axes.
- **Stage 8:** independent CF4 / EDD environment replication attempt.
- **Stage 9:** independent published local-void replication.

The strongest claims in this paper do not rely on any one stage in isolation. They rely on the structure of the chain as a whole.

4 Results I: Detection of a Low-Redshift Hubble Dipole

4.1 Reference dipole

The reference sample over $0.01 \leq z_{\mathrm{CMB}} \leq 0.15$ yields

$$|\mathbf{b}| = 0.080799 \text{ mag}, \tag{6}$$

$$(\mathrm{RA}, \mathrm{Dec}) = (23.331^\circ, 37.617^\circ). \tag{7}$$

The residual-dipole fit beat the corresponding shuffle null distribution. The observed dipole amplitude exceeded the strongest member of the null ensemble used in the reference fit.

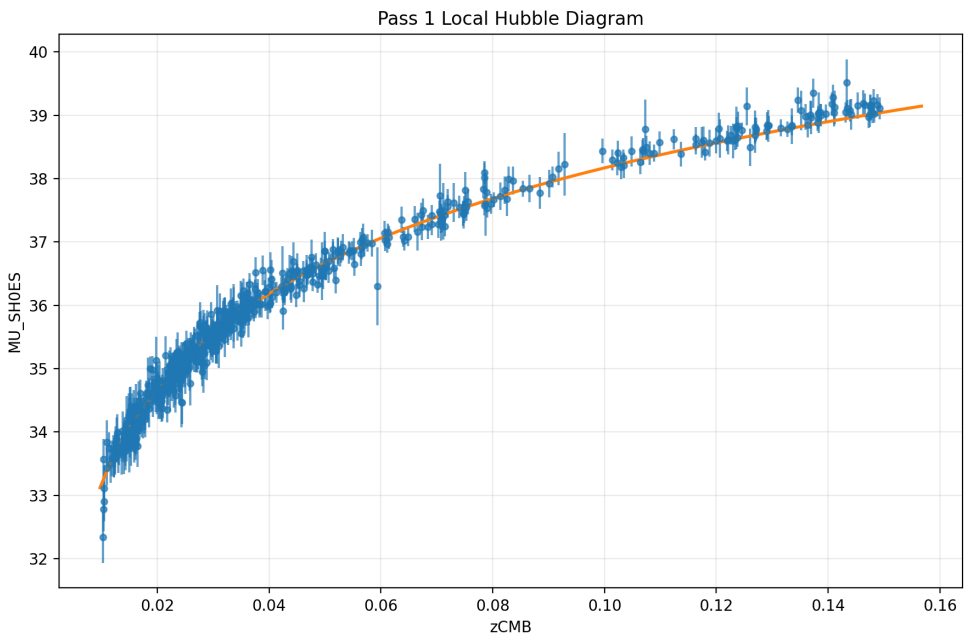

Figure 1: Reference low-redshift Hubble diagram for the cleaned Pantheon+SH0ES sample.

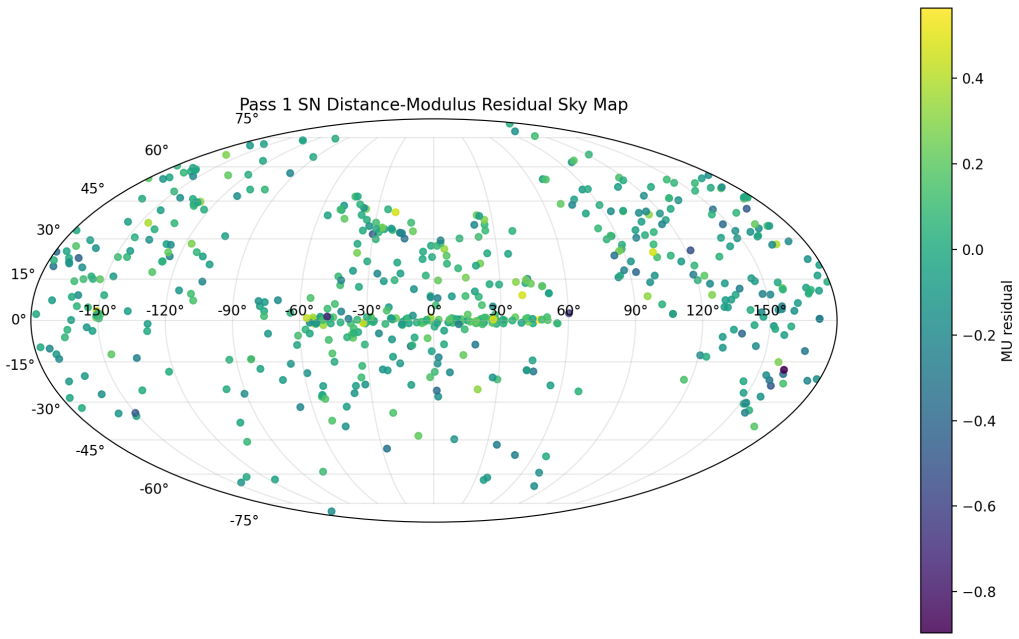

Figure 2: All-sky map of first-pass distance-modulus residuals. The map already suggests broad large-angle structure rather than purely uncorrelated scatter.

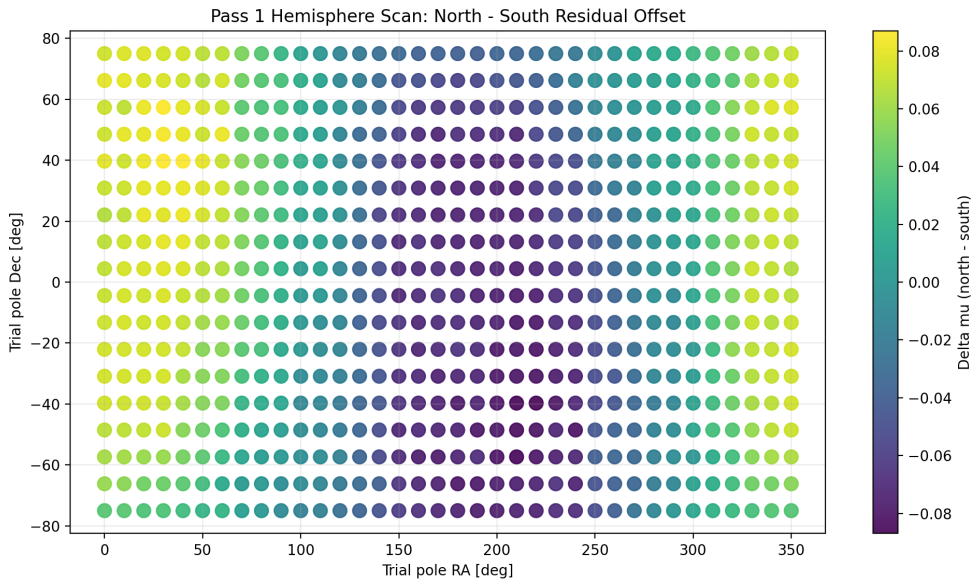

Figure 3: First-pass hemisphere scan. The smooth large-angle structure motivates direct dipole fitting rather than reliance on a single best hemisphere split.

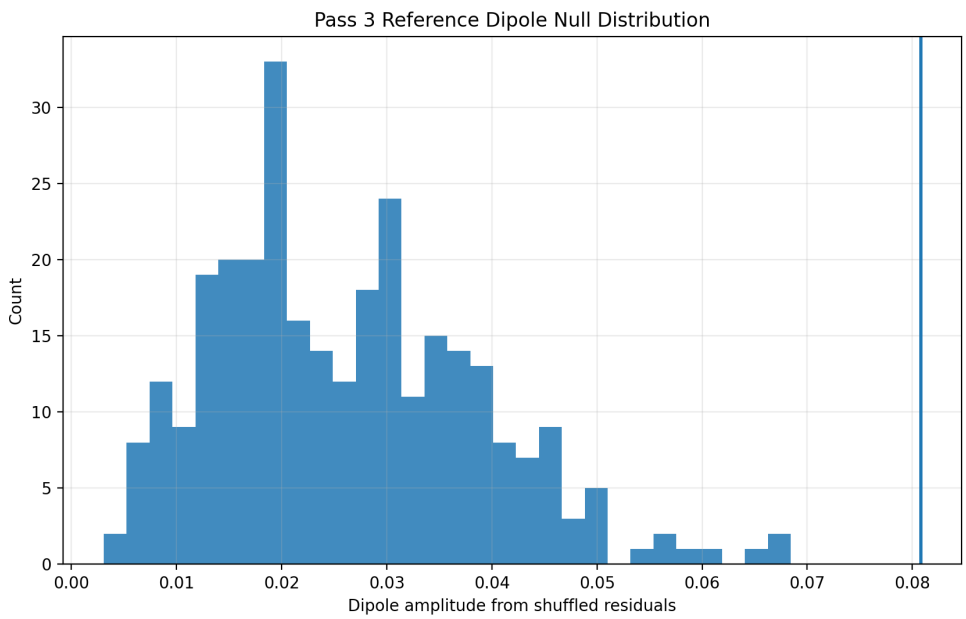

Figure 4: Reference dipole amplitude compared with the residual-shuffle null distribution.

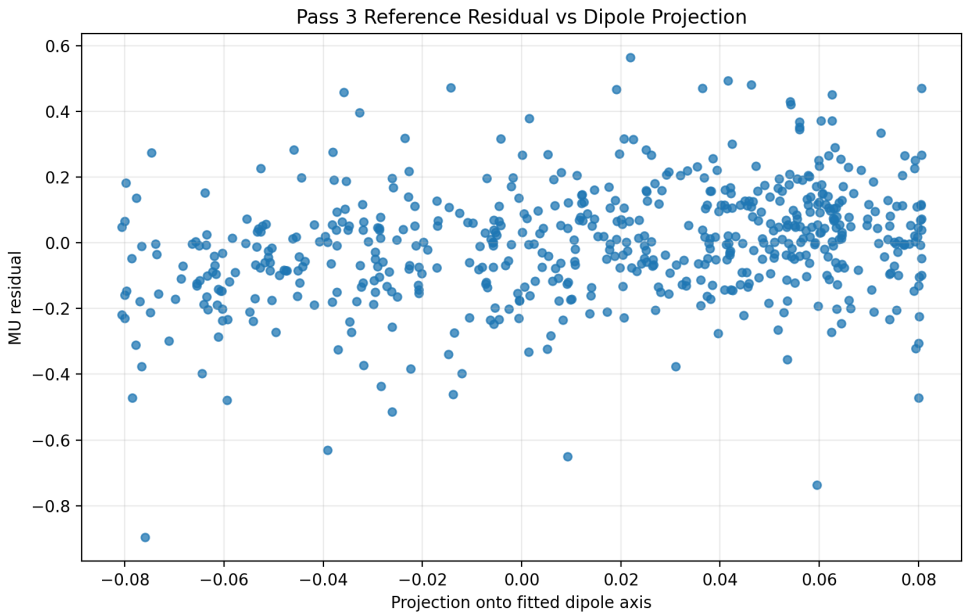

Figure 5: Reference residual versus fitted dipole projection. The trend is broad rather than razor-thin, but clearly directional rather than random.

4.2 Shell and cumulative stability

The radial behaviour is not completely uniform. The lowest shell, 0.01 to 0.03, carries a particularly clear dipolar component aligned within about 7° of the reference axis. Cumulative shells from $z_{\max} = 0.05$ to 0.12 remain relatively stable in both dipole amplitude and axis drift, while the highest shell, 0.10 to 0.15, appears noisier and differently oriented.

This matters for interpretation. The data do not support the claim that every shell independently

carries the same rigid dipole. What they do support is a coherent low-redshift directional component that dominates the cumulative sample and is strongest in the nearby universe.

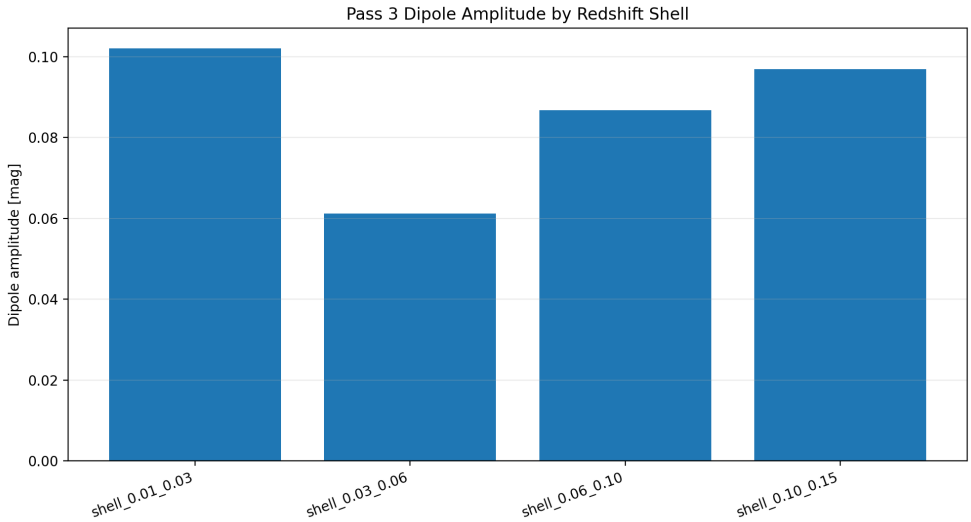

Figure 6: Dipole amplitude by fixed redshift shell.

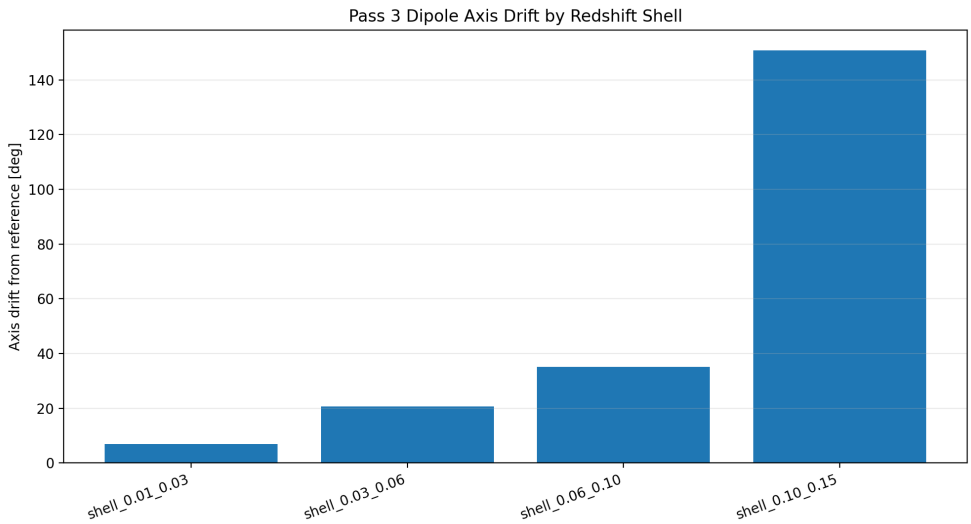

Figure 7: Dipole axis drift by fixed redshift shell relative to the reference axis.

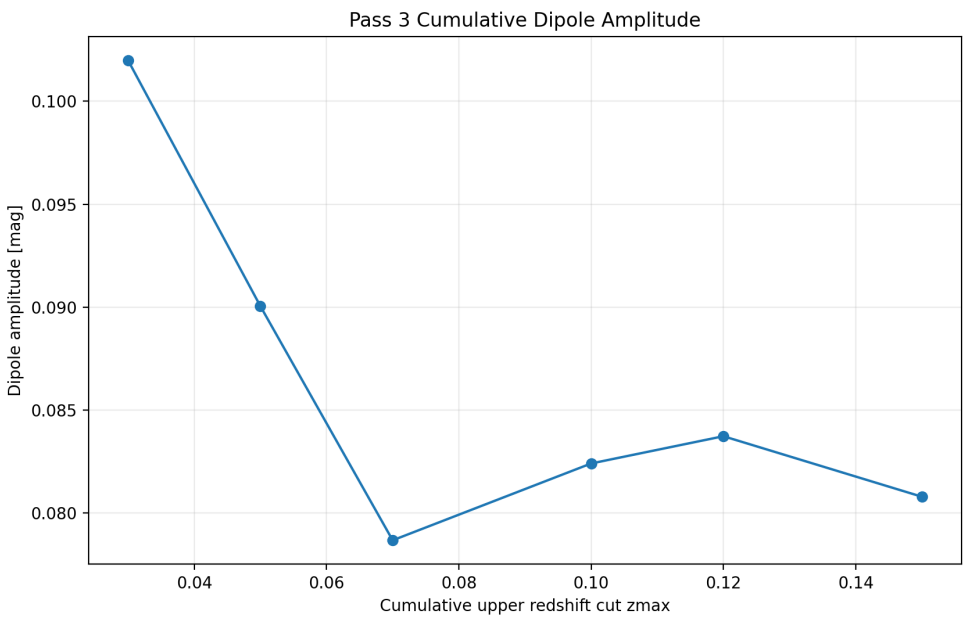

Figure 8: Cumulative dipole amplitude versus upper redshift cut.

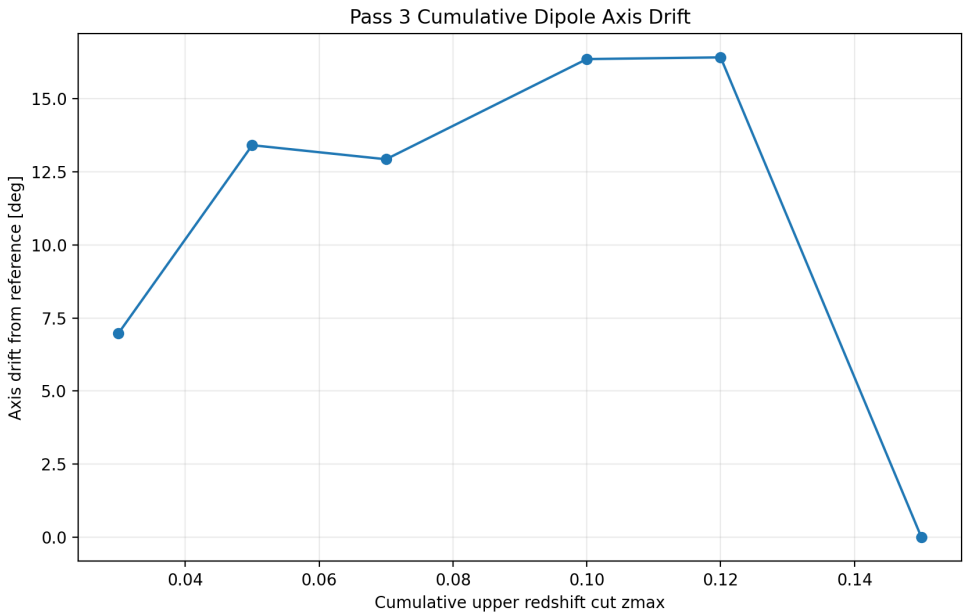

Figure 9: Cumulative dipole axis drift versus upper redshift cut.

5 Results II: Custom Environment and Stratification Field

5.1 Galaxy-based stratification result

Using the custom galaxy field, the strongest projection-side environment statistics are

$$\mathrm{corr}(\text{projection}, \text{shell}) = -0.489, \tag{8}$$

$$\mathrm{corr}(\text{projection}, \text{foreground}) = -0.274, \tag{9}$$

$$\mathrm{corr}(\text{projection}, \text{composite stratification}) = +0.469. \tag{10}$$

The residual-side couplings are weaker but still informative:

$$\mathrm{corr}(\text{residual}, \text{shell}) = -0.194, \tag{11}$$

$$\mathrm{corr}(\text{residual}, \text{foreground}) = -0.030, \tag{12}$$

$$\mathrm{corr}(\text{residual}, \text{composite stratification}) = +0.142. \tag{13}$$

All three projection-side null probabilities were effectively zero in the environment-shuffle ensemble. The shell and composite-stratification residual-side links also beat their nulls, whereas the foreground-only residual statistic did not.

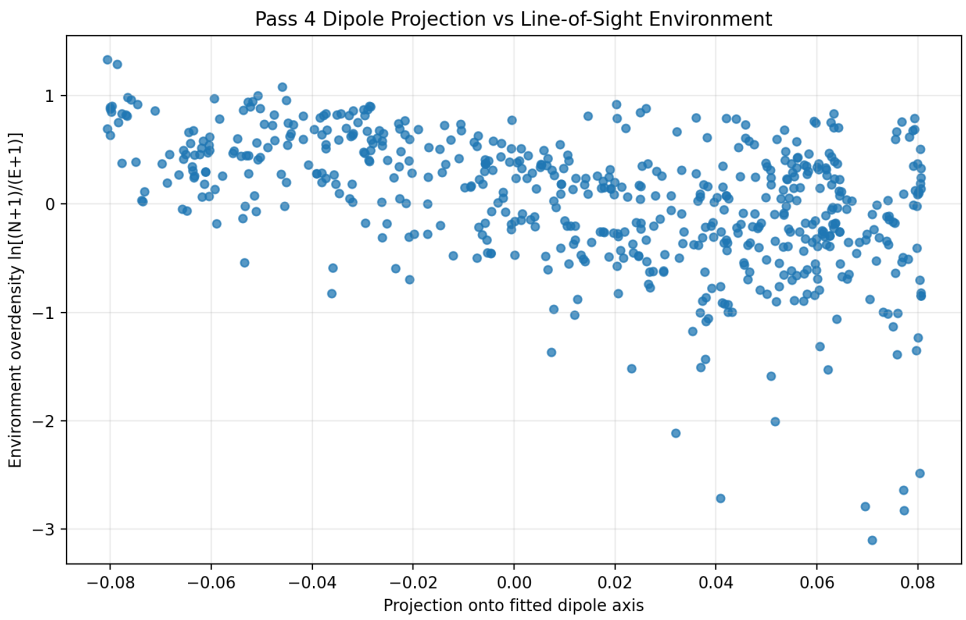

Figure 10: Reference dipole projection versus line-of-sight overdensity proxy from the custom galaxy field.

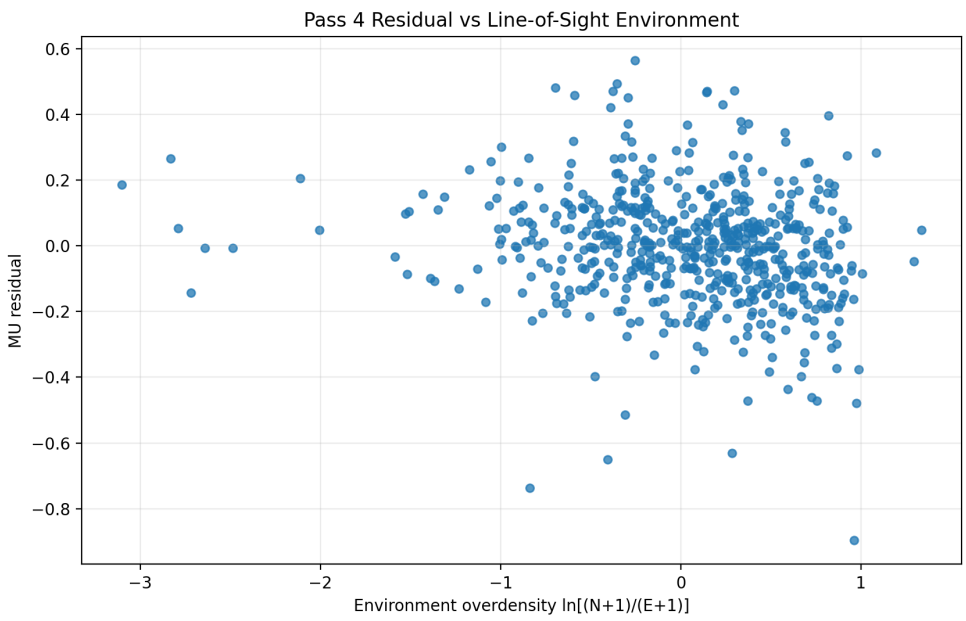

Figure 11: Reference residual versus custom environment proxy.

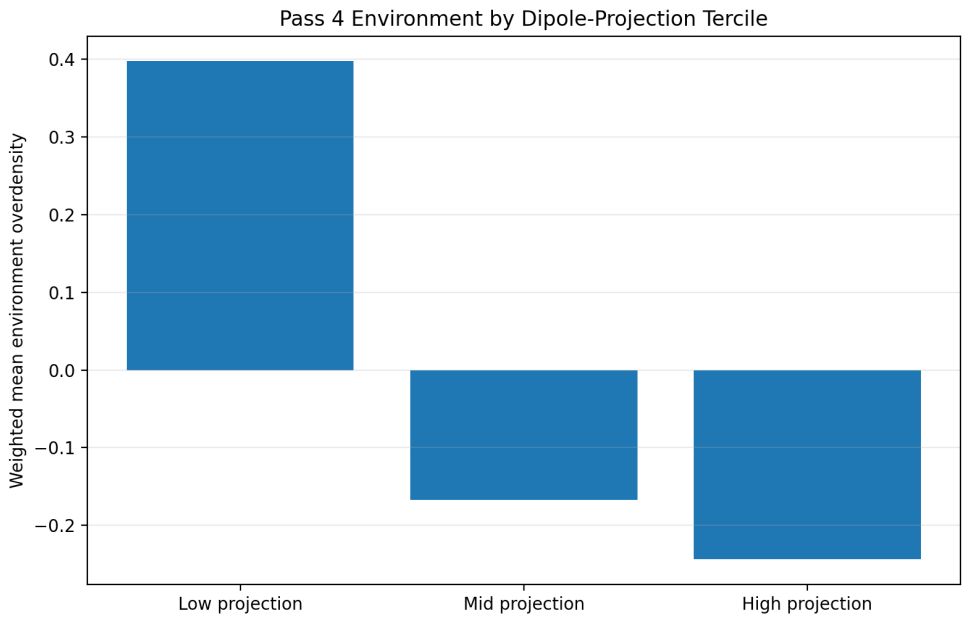

Figure 12: Weighted mean environment metric across low-, mid-, and high-projection terciles. Opposite sides of the dipole sample measurably different environments.

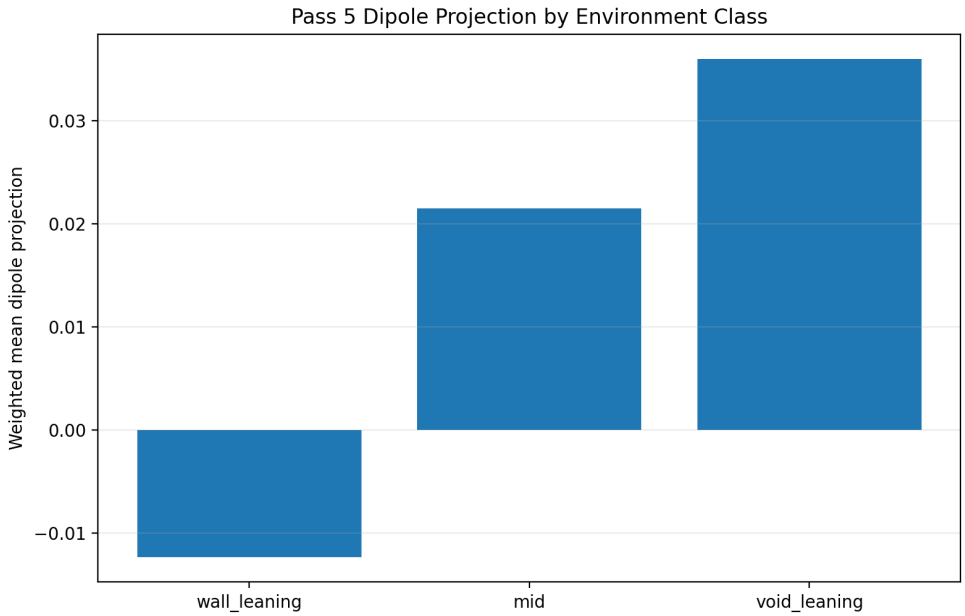

Figure 13: Dipole projection by composite environment class from the custom galaxy stratification field.

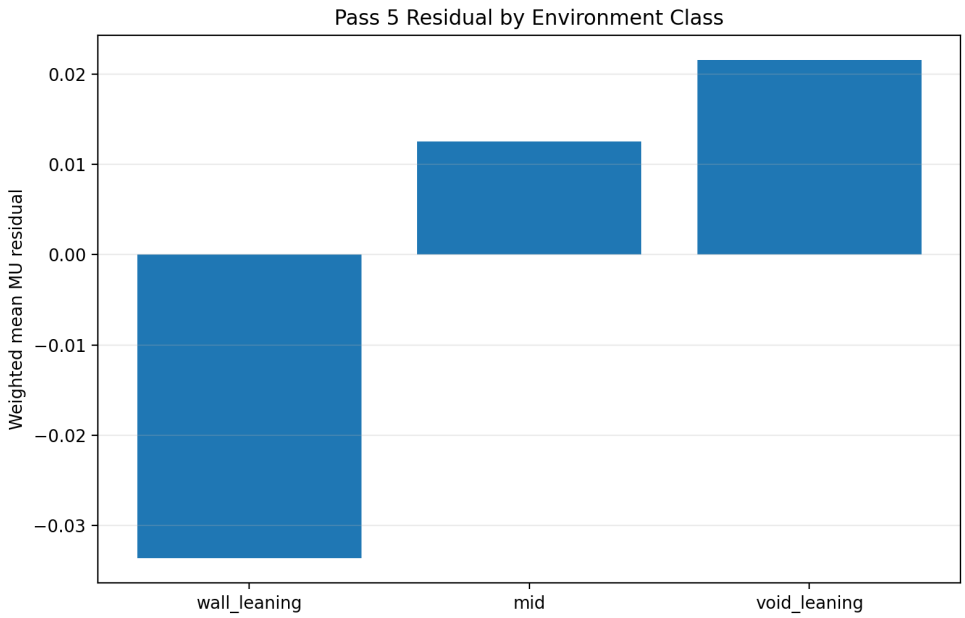

Figure 14: Residual by composite environment class from the custom galaxy stratification field.

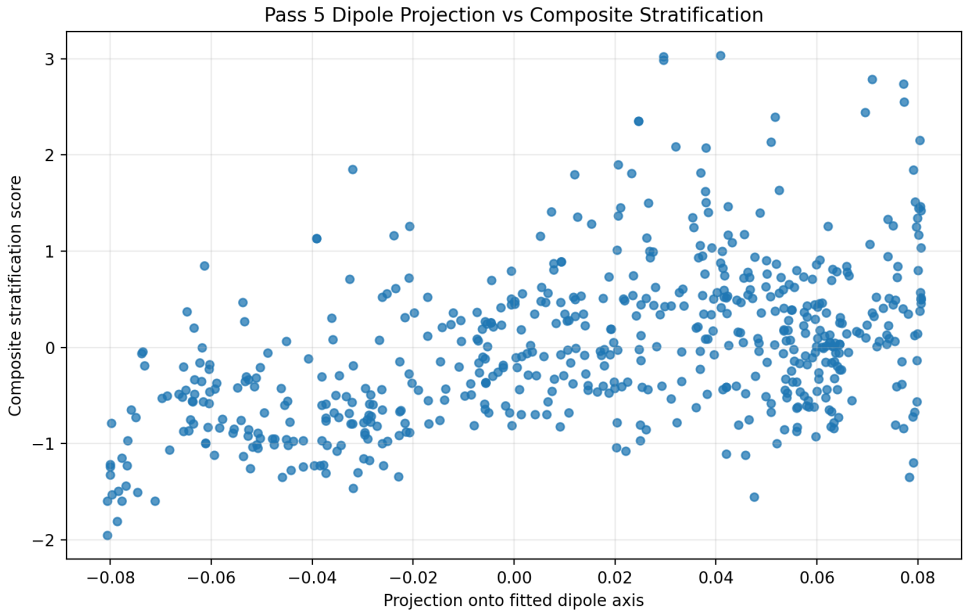

Figure 15: Projection versus composite stratification score.

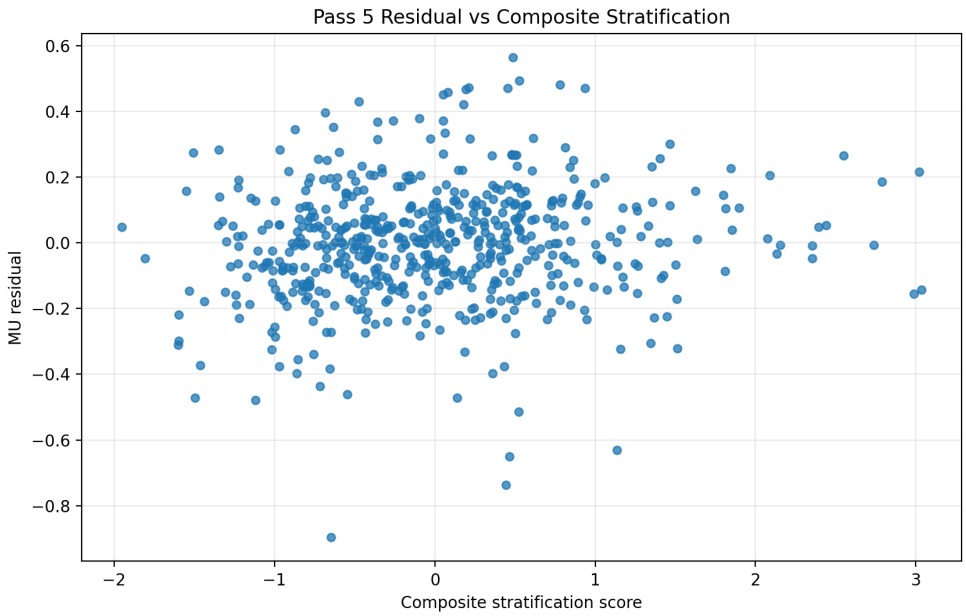

Figure 16: Residual versus composite stratification score.

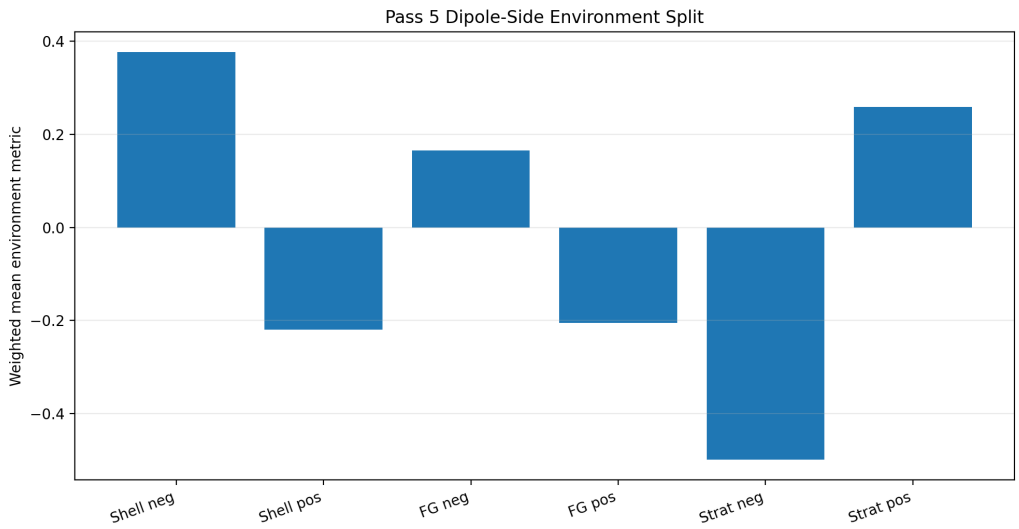

Figure 17: Dipole-side split in environment metrics from the custom stratification analysis.

5.2 Interpretation

The custom environment result is not simply that one side of the sky looks different from the other. It is that the fitted Hubble dipole partitions the supernova sample into sectors that differ in line-of-sight environment in a consistent and redshift-respecting way. This is the point where the result first becomes environment-shaped rather than merely directional.

6 Results III: Falsification and Axis Specificity

6.1 What survives

Against redshift-matched and radial-scramble nulls, the custom stratification link survives.

For the custom stratification field, the absolute null probability for the projection–stratification correlation under the redshift-matched null is 0.0. The corresponding redshift-matched null probability for the residual–stratification correlation is 0.0025. Under the radial-scramble null, the residual–stratification correlation again gives an absolute null probability of 0.0025.

The projection–stratification link also remains stable when the low-redshift floor is raised through $z_{\min} = 0.02$, 0.03, and 0.04, although the dipole amplitude itself weakens as the nearest supernovae are removed.

6.2 What does not survive

A crude sky-rotation test does not preserve uniqueness of the simple global environment correlation.

For the custom stratification field, the absolute null probability for the projection–stratification correlation under the sky-rotation null is 0.8675. The corresponding sky-rotation null probability for the residual–stratification correlation is 1.0.

This is a crucial limitation. It means the result cannot honestly be framed as "the fitted dipole axis is uniquely selected by the environment field" if one judges specialness only by simple global correlation.

6.3 Axis-specific clarification

Control-axis tests resolve this partially. When the fitted dipole is compared against many control axes drawn from the same sky mask, the simple projection–stratification correlation is not uniquely extreme. Under the mask-matched control-axis test, the absolute null probability for the projection–stratification correlation is 0.215.

However, the fitted axis is unusually strong in direct sector separation. Under the same mask-matched control-axis family, the absolute null probability for the matched stratification side split is 0.085, while the corresponding value for the matched residual side split is 0.021667.

In percentile terms, the fitted axis is around the 91.5th percentile for matched stratification side split and the 97.8th percentile for matched residual side split relative to mask-matched control axes.

So the strongest honest statement is not that the dipole axis is uniquely best by raw correlation, but that it is unusually effective at partitioning the sample into opposite sectors with different environment and residual behaviour.

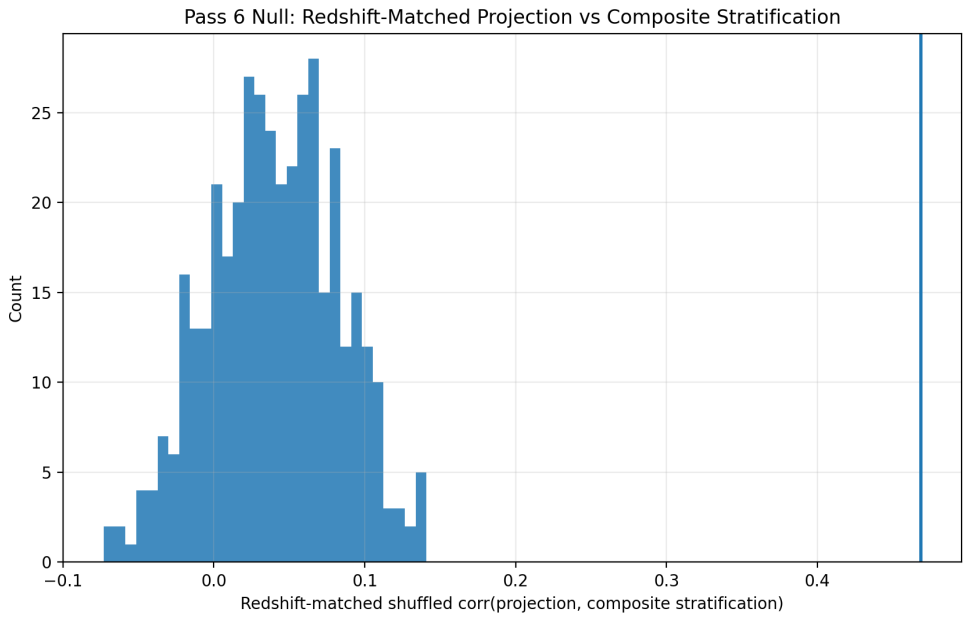

Figure 18: Redshift-matched null distribution for the projection–stratification statistic.

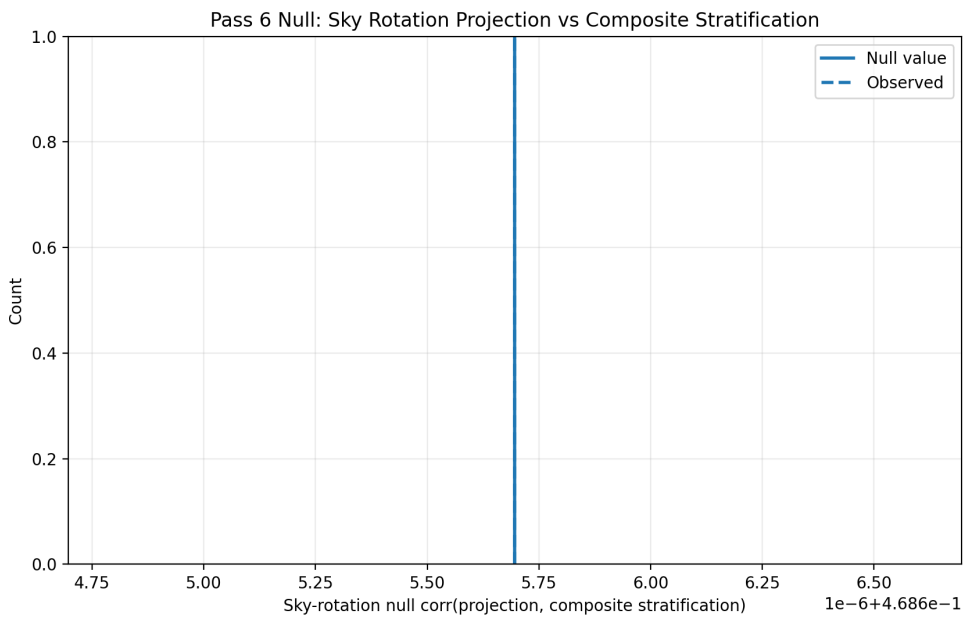

Figure 19: Sky-rotation null distribution for the projection–stratification statistic. The observed value is not unique relative to this broad angular control family.

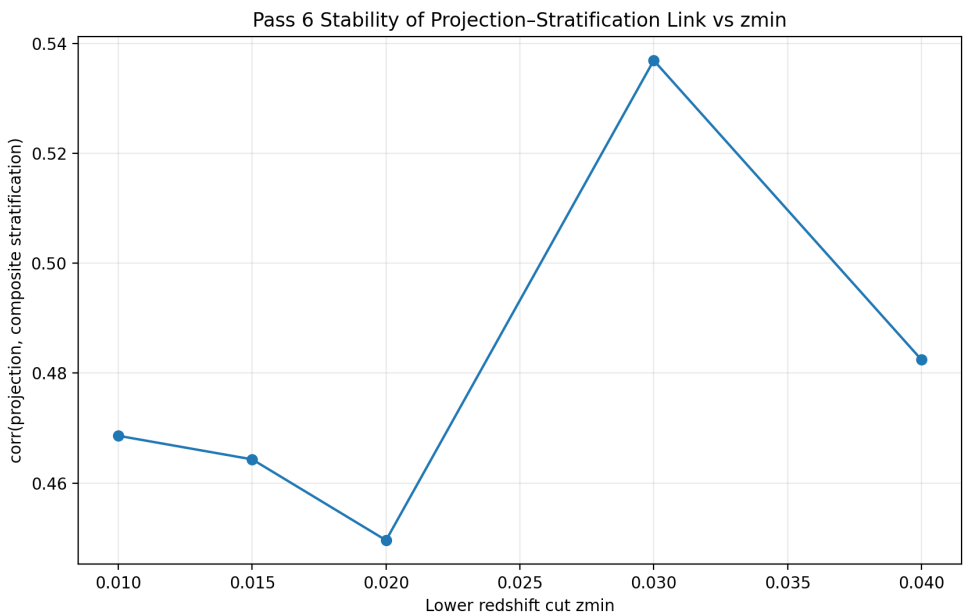

Figure 20: Projection–stratification stability versus increasing low-redshift floor.

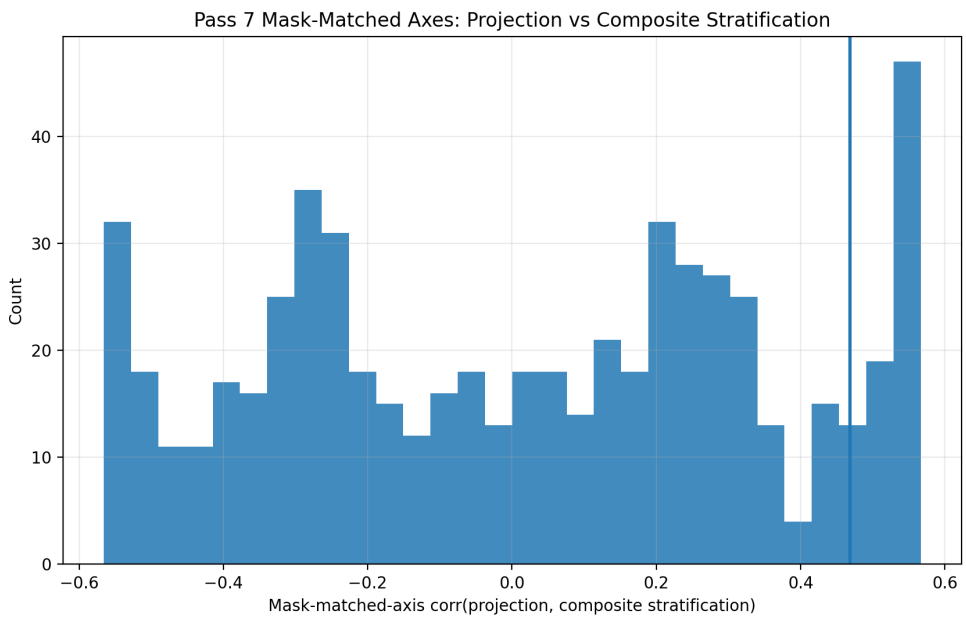

Figure 21: Mask-matched control-axis distribution of the projection–stratification statistic.

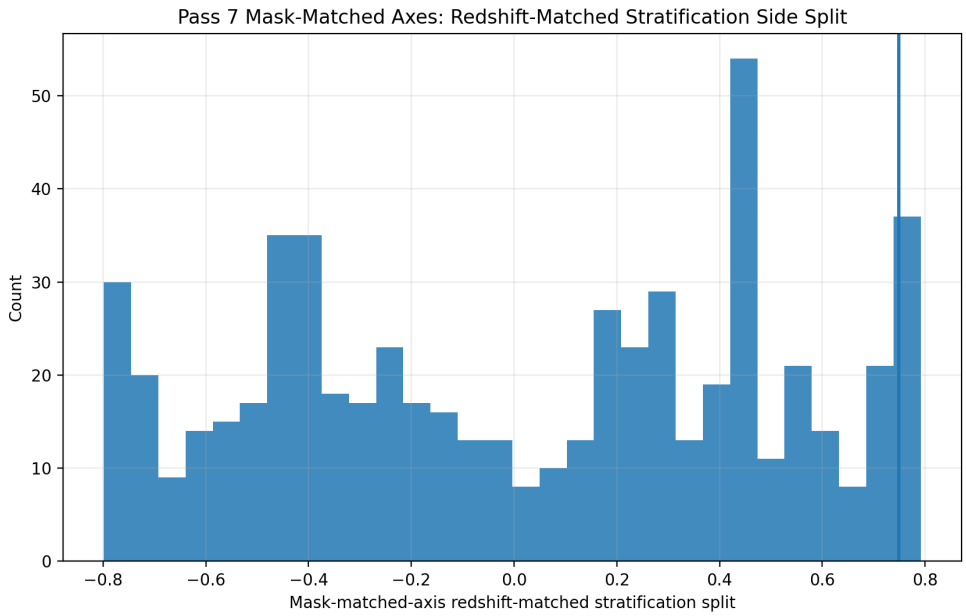

Figure 22: Mask-matched control-axis distribution for the redshift-matched stratification side split.

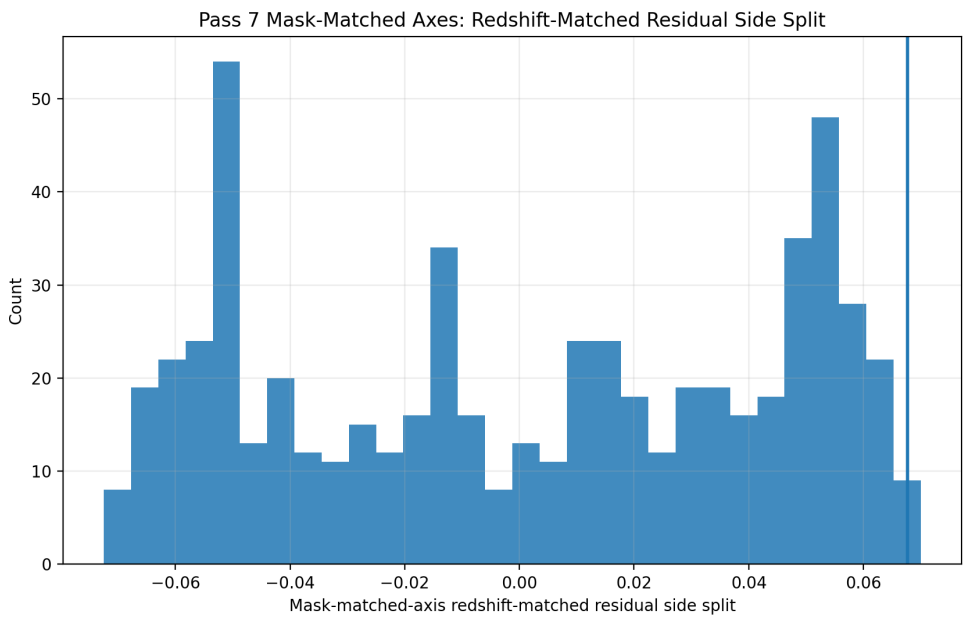

Figure 23: Mask-matched control-axis distribution for the redshift-matched residual side split.

7 Results IV: Independent Replication Attempts

7.1 CF4 / EDD non-replication

We deliberately retain the failed CF4 / EDD attempt because it is part of the honest evidentiary chain. Using the CF4 / EDD table as an independent structure field, the main environment correlations become weak:

$$\mathrm{corr}(\mathrm{projection}, \mathrm{CF4\ strat}) = -0.057, \tag{14}$$

$$\mathrm{corr}(\mathrm{residual}, \mathrm{CF4\ strat}) = -0.001. \tag{15}$$

The associated null probabilities are not compelling. The direct side-split metrics show mild structure, but the stronger matched-sector versions do not replicate.

This suggests that the CF4 proxy, at least in the simple line-of-sight implementation used here, is not a suitable independent environment field for the present question.

7.2 Published-void replication

The crucial independent replication comes from the published LocalVoids catalogue. In this stage, each supernova sightline is tested against nearby published voids using angular overlap and radial consistency in velocity space. This yields a published-void overlap score and a corresponding void-conditioned stratification metric.

The resulting statistics are strong:

$$\mathrm{corr}(\text{projection}, \text{void hits}) = -0.3738, \tag{16}$$
$$\mathrm{corr}(\text{projection}, \text{void overlap}) = -0.3399, \tag{17}$$
$$\mathrm{corr}(\text{residual}, \text{void hits}) = -0.2375, \tag{18}$$
$$\mathrm{corr}(\text{residual}, \text{void overlap}) = -0.2065. \tag{19}$$

Projection-side null probabilities are effectively zero:

$$p(|\mathrm{corr}(\text{projection}, \text{void hits})|) = 0.0, \tag{20}$$
$$p(|\mathrm{corr}(\text{projection}, \text{void overlap})|) = 0.0. \tag{21}$$

The direct and matched sector splits are equally important:

$$\Delta^{(+)-(-)}_{\text{void hits}} = -0.9761, \qquad p = 0.0, \tag{22}$$
$$\Delta^{(+)-(-)}_{\text{void overlap}} = -0.7015, \qquad p = 0.0, \tag{23}$$
$$\Delta^{(+)-(-)}_{\text{matched, void hits}} = -0.5912, \qquad p = 0.0, \tag{24}$$
$$\Delta^{(+)-(-)}_{\text{matched, void overlap}} = -0.4136, \qquad p = 0.0. \tag{25}$$

This is the strongest external confirmation in the paper. The sign convention of the provisional class labels can be flipped depending on how one names the class centroids, but the sector-based statistics are invariant and are the real point of the test.

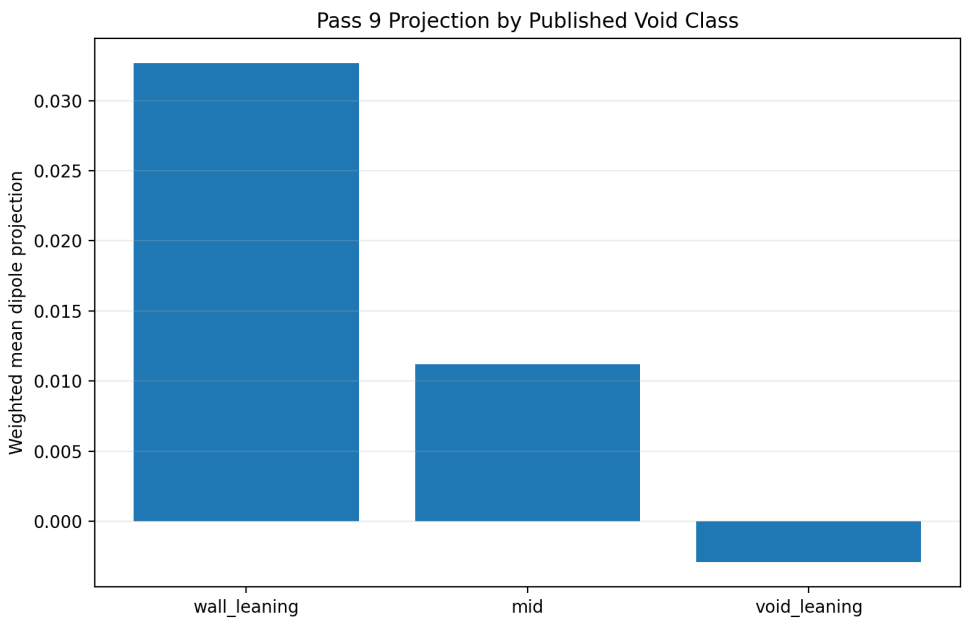

Figure 24: Dipole projection by class derived from the published LocalVoids overlap field.

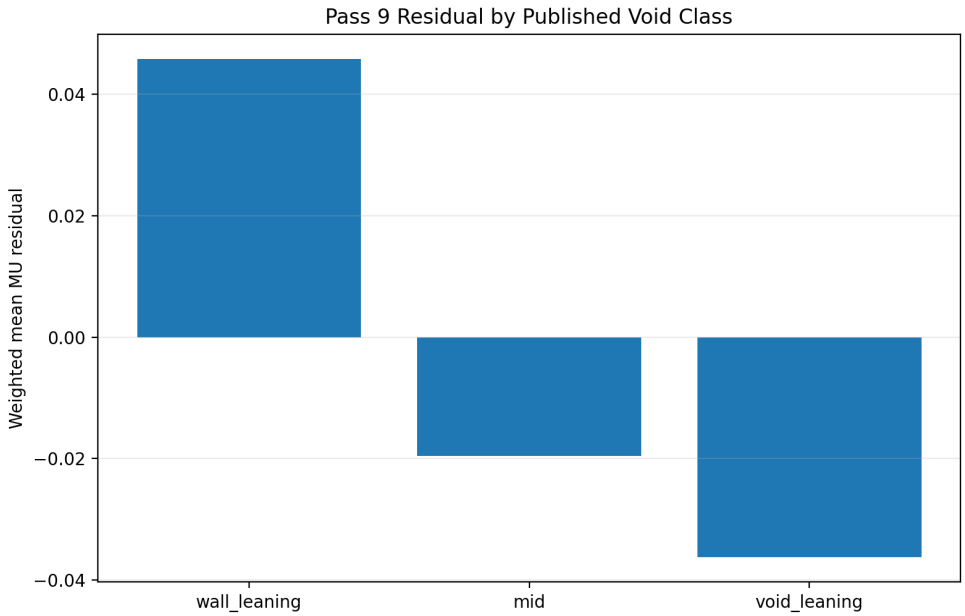

Figure 25: Residual by class derived from the published LocalVoids overlap field.

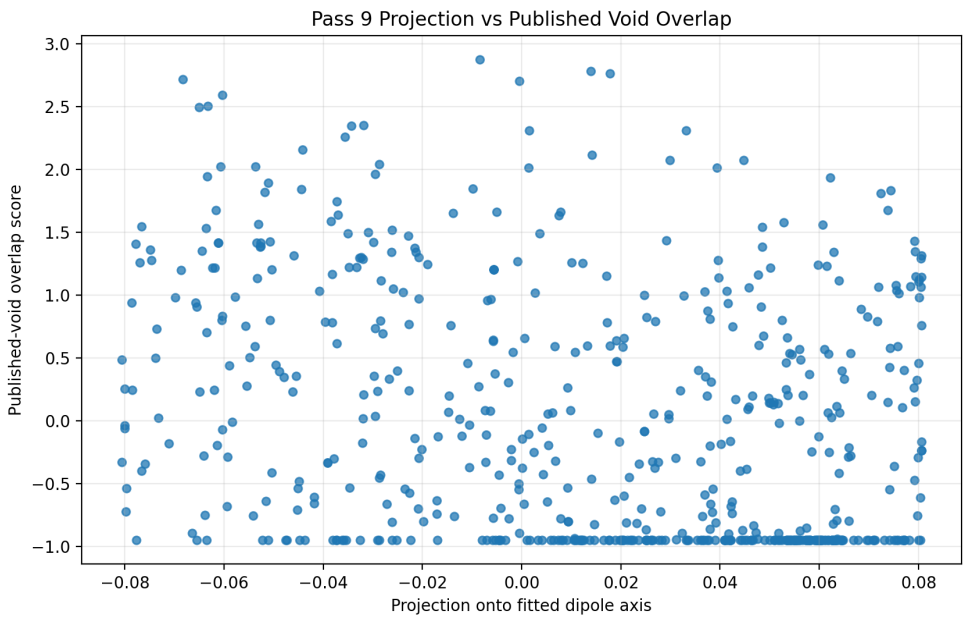

Figure 26: Projection versus published-void overlap score.

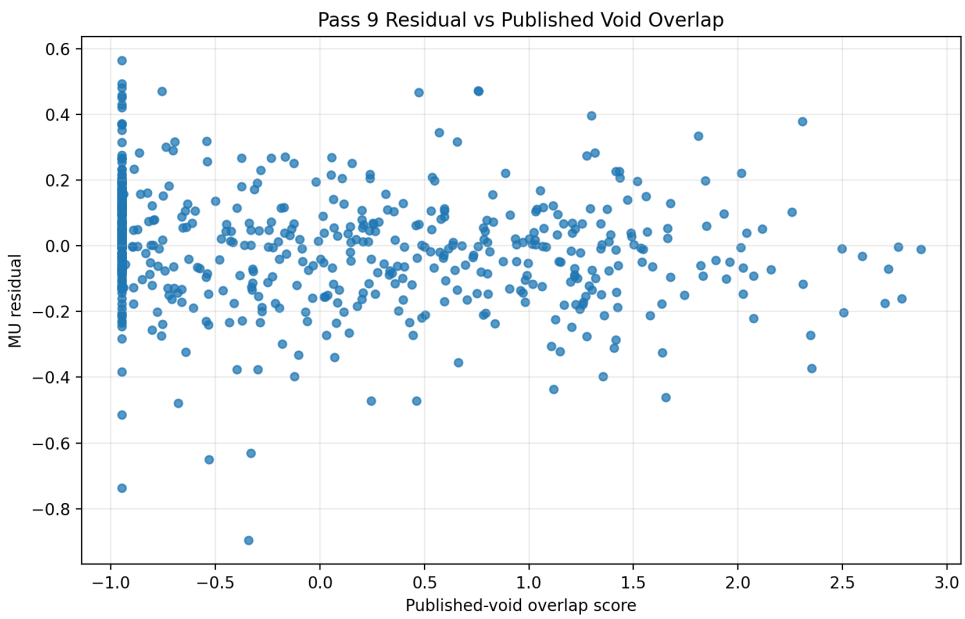

Figure 27: Residual versus published-void overlap score.

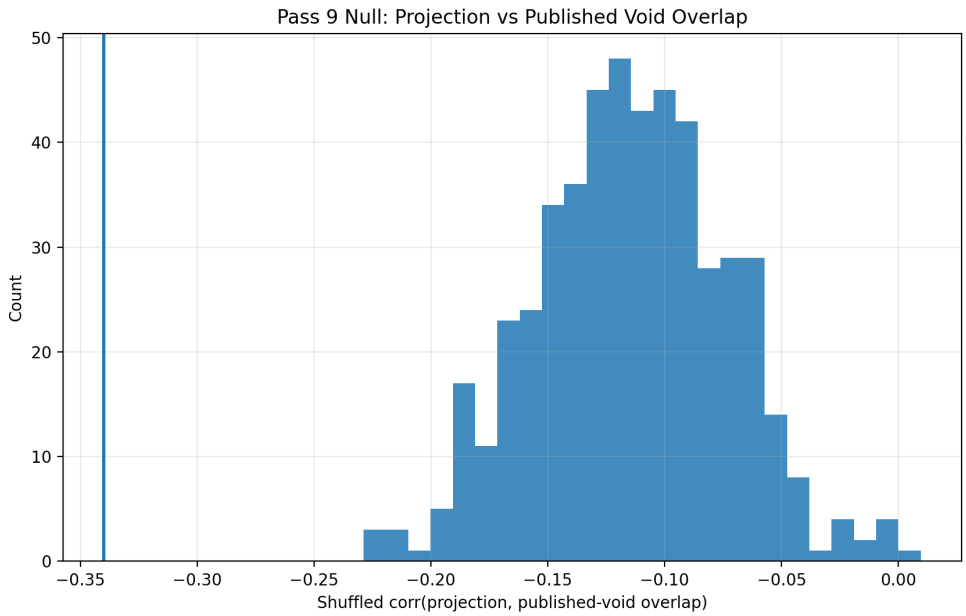

Figure 28: Null distribution for the projection–published-void overlap statistic.

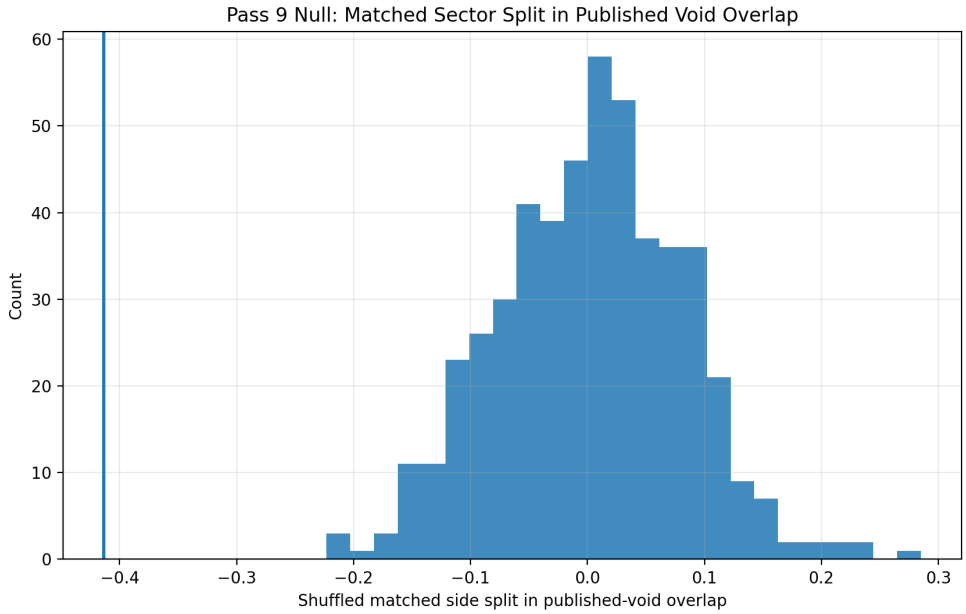

Figure 29: Null distribution for the redshift-matched sector split in published-void overlap.

8 Compact Quantitative Summary

Table 1 summarises the main quantitative results across the most important stages.

Table 1: Key quantitative results across the analysis chain.

Stage / test	Quantity	Value	Interpretation
Reference dipole	Dipole amplitude	0.0808 mag	Real low-z dipolar residual field
Reference dipole axis	Best-fit axis (RA, Dec)	$(23.3°, 37.6°)$	Reference large-angle direction
Custom galaxy field	Projection–stratification correlation	+0.469	Strong custom environment coupling
Custom galaxy field	Residual–stratification correlation	+0.142	Weaker but nontrivial residual coupling
Redshift-matched null	Null probability for projection–stratification correlation	0.0	Not a trivial redshift-distribution artefact
Sky-rotation null	Null probability for projection–stratification correlation	0.8675	Raw correlation not uniquely special under crude angular controls
Mask-matched control axes	Null probability for matched residual side split	0.0217	Fitted axis unusually strong in matched residual sector separation
CF4 / EDD replication	Projection–CF4 stratification correlation	−0.057	No convincing independent replication
Published void replication	Projection–void-overlap correlation	−0.340	Strong independent projection-side replication
Published void replication	Null probability for matched void-overlap sector split	0.0	Strong independent matched sector replication

9 Discussion

9.1 What the data support

The strongest statement supported by the present analysis is:

> Low-redshift Hubble residuals define a significant dipolar field whose opposite sky sectors are systematically separated in line-of-sight environment, and that sector-based environmental separation replicates against a published local-void catalogue.

This statement is stronger and more honest than several tempting alternatives. In particular, we do not claim that the fitted dipole axis is uniquely selected by the environment field under all controls. Stages 6 and 7 show that such an overstatement would not be warranted.

What the evidence does show is that once the actual Hubble dipole is fit from the supernova residual field, the resulting positive and negative sectors are not environmentally interchangeable. They differ in galaxy-based stratification and, crucially, in overlap with a published local-void catalogue, even when redshift matching is enforced.

9.2 What the data do not yet support

The current results do not by themselves prove any one specific physical mechanism. They do not show that the effect is impossible to explain within a refined local-structure or flow interpretation. They also do not show that every independent structure field must recover the same effect. The CF4 / EDD non-replication is a clear reminder of that last point.

The results therefore sit in an intermediate but important position:

- stronger than a purely exploratory directional anomaly,
- weaker than a complete mechanism-level demonstration,
- but strong enough to justify formal presentation and external review.

9.3 Implications for late-time expansion inference

These results are consistent with a picture in which local expansion inference is conditioned by how different lines of sight sample nearby structure, especially voids and walls. At minimum, they imply that treating the low-redshift Hubble relation as purely isotropic residual scatter is incomplete. At maximum, they support the broader claim that late-time expansion inference should be analysed as a direction- and environment-conditioned field rather than as a single scalar summary.

10 Limitations

Several limitations should be stated explicitly.

First, the baseline Hubble model is intentionally simple. It is adequate for residual-field discovery, but a final production analysis should include a more careful cosmographic baseline.

Second, the custom galaxy stratification field is a constructed proxy. It is useful and internally robust, but it is not a published density reconstruction.

Third, the CF4 / EDD replication attempt did not independently confirm the result. That non-replication is retained because it is informative and because it sharpens the argument that explicit void structure is more relevant for the present question than the particular CF4 proxy used here.

Fourth, the strongest claims in this paper are sector-based and matched-sector based, not uniqueness claims about one raw axis-correlation statistic.

11 Conclusion

We have presented a staged empirical analysis of the local Hubble relation in which the main question is whether low-redshift Hubble residuals are both directional and environment-conditioned.

The answer is yes.

A significant low-redshift dipolar residual field is present in the Pantheon+SH0ES sample. That field is not merely directional. It is structured by line-of-sight environment, and the resulting

sector separation survives a substantial chain of internal checks. The strongest custom-field evidence survives redshift-matched and radial-scramble tests, though not the most naive sky-rotation uniqueness criterion. Control-axis comparisons show that the fitted dipole is unusually effective at producing matched sector separation in environment and residual behaviour. Most importantly, the key sector-based result replicates against a published local-void catalogue.

The most defensible final statement is therefore:

> The observed local Hubble dipole defines opposite sky sectors that are unusually separated in both line-of-sight stratification and redshift-matched Hubble residual behaviour, and this sector-based environmental split is independently replicated by a published local-void catalogue.

That is sufficient to move beyond an exploratory anomaly report and into a formal evidence paper.

References

[1] Brout, D., Scolnic, D., Popovic, B., et al. (2022). *The Pantheon+ Analysis: Cosmological Constraints.* The Astrophysical Journal, 938, 110. arXiv:2202.04077.

[2] Scolnic, D., Brout, D., Carr, A., et al. (2022). *The Pantheon+ Analysis: The Full Dataset and Light-Curve Release.* The Astrophysical Journal, 938, 113. arXiv:2112.03863.

[3] Riess, A. G., Yuan, W., Macri, L. M., et al. (2022). *A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s^{-1} Mpc^{-1} Uncertainty from the Hubble Space Telescope and the SH0ES Team.* The Astrophysical Journal Letters, 934, L7. arXiv:2112.04510.

[4] Planck Collaboration, Aghanim, N., Akrami, Y., et al. (2020). *Planck 2018 Results. VIII. Gravitational Lensing.* Astronomy & Astrophysics, 641, A8. arXiv:1807.06210.

[5] Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves.* The Astronomical Journal, 152, 157. arXiv:1606.09251.

[6] McGaugh, S. S., Lelli, F., & Schombert, J. M. (2016). *Radial Acceleration Relation in Rotationally Supported Galaxies.* Physical Review Letters, 117, 201101. arXiv:1609.05917.

[7] Malandrino, R., Jasche, J., Lavaux, G., & Wandelt, B. D. (2025). *A Bayesian Catalog of 100 High-Significance Voids in the Local Universe.* arXiv:2507.06866.

[8] Hu, W., & Okamoto, T. (2002). *Mass Reconstruction with CMB Polarization.* The Astrophysical Journal, 574, 566–574. arXiv:astro-ph/0111606.

[9] Lewis, A., & Challinor, A. (2006). *Weak Gravitational Lensing of the CMB.* Physics Reports, 429, 1–65. arXiv:astro-ph/0601594.

[10] Darke, James Alan. (2026). *Little Red Dots as Early Closure Nodes.* Zenodo Record: 18899599.